Thermal Equilibrium

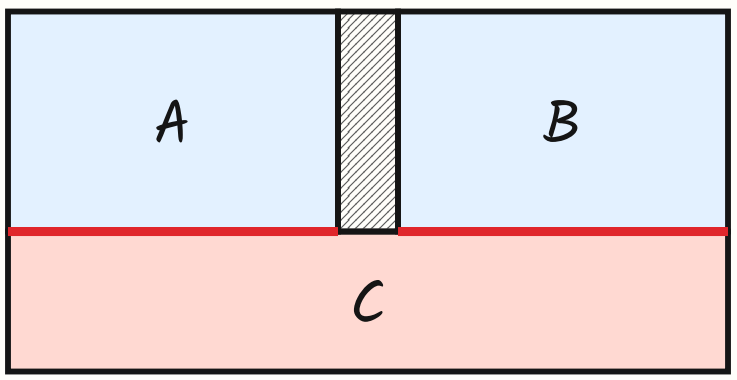
Class 11 · IDEAL gas, n moles FIXED (closed system) unless stated · W = work done BY the gas · quasi-static paths on P–V graphs · $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$ · T in kelvin

1 System, surroundings, walls

- SYSTEM: the gas studied; SURROUNDINGS: the rest. ADIABATIC wall: no heat passes; DIATHERMIC wall: heat passes.
- State variables (EQUILIBRIUM only): P, V, T, U; intensive P, T; extensive V, U, n.

2 Zeroth law of thermodynamics

adiabatic wall between A and B



red: diathermic walls (A–C, B–C)

$$T_A = T_C \text{ and } T_B = T_C$$

$$\Rightarrow\ T_A = T_B$$

A, B EACH in thermal equilibrium with C ⇒ A, B in thermal equilibrium with each other

- Thermal equilibrium: no NET heat flow through a diathermic contact; the EQUAL property is TEMPERATURE (basis of thermometers).

3 Equation of state

$$PV = nRT$$

IDEAL gas in EQUILIBRIUM · P in Pa, V in m^3, T in K, $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$ · one state = one point on P–V

Syllabus Checklist

JEE Main + JEE Advanced syllabus + standard results

1 JEE Main + JEE Advanced syllabus

- ✓ Thermal equilibrium, zeroth law, temperature — p1
- ✓ Heat, work and internal energy — p3
- ✓ First law of thermodynamics, sign convention — p4
- ✓ Quasi-static processes, work = area under P–V — p5, p6
- ✓ Isochoric and isobaric processes — p7
- ✓ Isothermal and adiabatic processes — p8, p9
- ✓ Slopes of isothermal and adiabatic curves; bulk modulus of gases — p10
- ✓ Specific heats C_V, C_P; Mayer's relation — p12
- ✓ γ for monatomic, diatomic, polyatomic gases — p13
- ✓ Cyclic processes — p17
- ✓ Heat engines, efficiency; refrigerator, COP — p19, p20
- ✓ Carnot engine and its efficiency; Carnot's theorem — p21, p22
- ✓ Second law: Kelvin–Planck, Clausius — p22
- ✓ Reversible and irreversible processes — p23

2 Also covered: extra topics + standard results

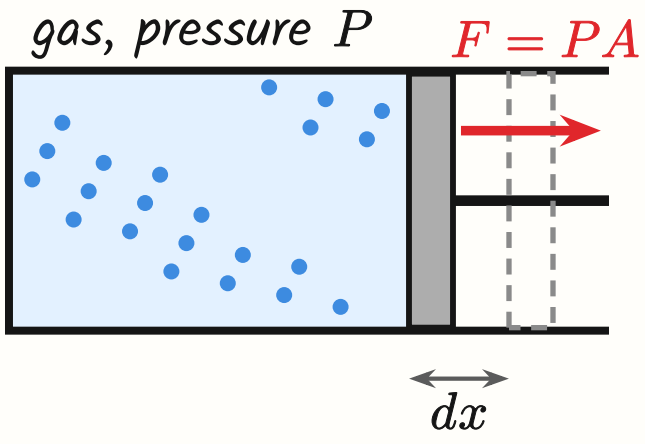
- ✓ Polytropic process $PV^x = \text{const}$, its heat capacity — p11
- ✓ Mixtures of gases: C_V, γ_{mix}, mixing temperature — p14
- ✓ P–V, V–T, P–T graphs; converting a cycle — p15, p16
- ✓ Free expansion; constant external pressure — p18
- ✓ Work in isothermal, adiabatic, polytropic processes — p8, p9, p11
- ✓ Equipartition of energy, degrees of freedom — p13
- ✓ Heat pump; same machine as engine and refrigerator — p20
- ✓ Entropy basics, T–S diagram (JEE Advanced) — p24
- ✓ Process table, results, traps, solved questions — p25–p32

Heat, Work, Internal Energy

1 Heat Q

- Energy that flows ONLY because of a temperature difference. Not stored in a body; a PATH function.

2 Work done BY the gas



$$dW = PA\,dx = P\,dV$$

$$W = \int_{V_1}^{V_2} P\,dV$$

QUASI-STATIC (P = gas pressure) · expansion $W > 0$, compression $W < 0$

3 Internal energy U

$$\Delta U = nC_V \Delta T$$

IDEAL gas, ANY process (C_V constant) · U depends on T only (no intermolecular forces)

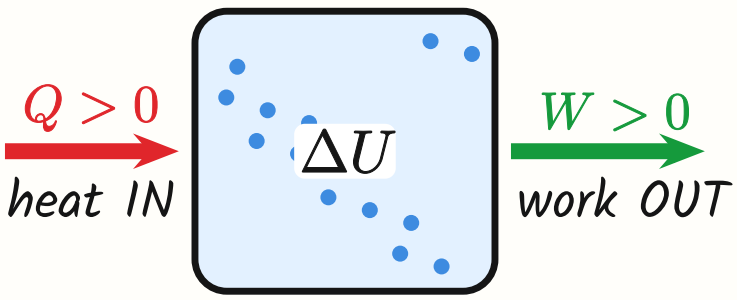
- U = total KE + PE of the molecules (centre-of-mass frame). A STATE function: ΔU depends only on the end states.

Golden points

★ Q and W depend on the PATH; ΔU does not. Heat and work are energy in TRANSIT, not contents.

First Law of Thermodynamics

1 Statement



$$Q = \Delta U + W$$

$$dQ = dU + PdV$$

energy conservation for the gas · W = work done BY the gas · second form: quasi-static

2 Sign convention (used on every page)

Quantity	POSITIVE when	NEGATIVE when
Q	heat GIVEN to the gas	heat taken OUT of the gas
W	gas EXPANDS (work BY it)	gas is COMPRESSED
ΔU	U rises (T rises, ideal gas)	U falls (T falls)

- Also written $\Delta U = Q + W_{on}$ with $W_{on} = -W$: the SAME law.

3 Worked signs

Example

Heat $500\,\mathrm{J}$ given, gas does $200\,\mathrm{J}$ of work: $\Delta U = 500 - 200 = 300\,\mathrm{J}$.
Compressed with $100\,\mathrm{J}$ of work done ON it, loses $40\,\mathrm{J}$ of heat:
$Q = -40$, $W = -100$, $\Delta U = -40 + 100 = +60\,\mathrm{J}$.

Golden points

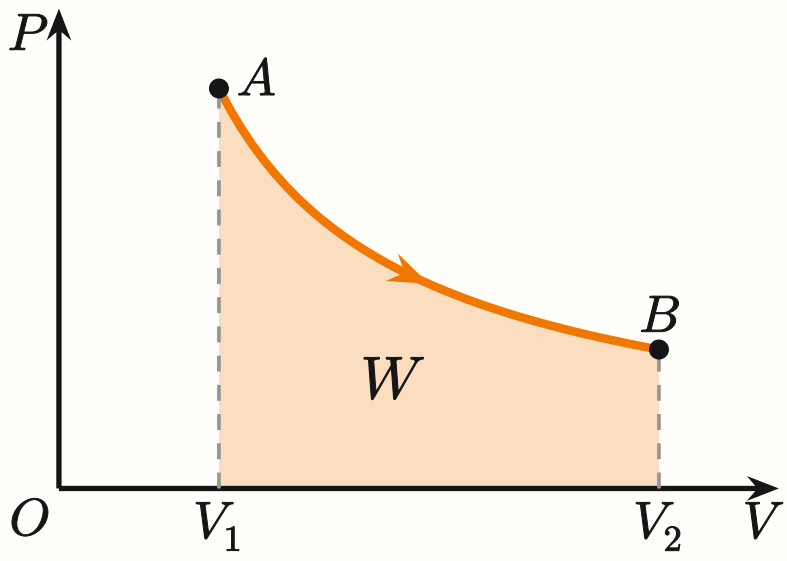
★ Valid for EVERY process between equilibrium states; it cannot tell the DIRECTION (second law can).

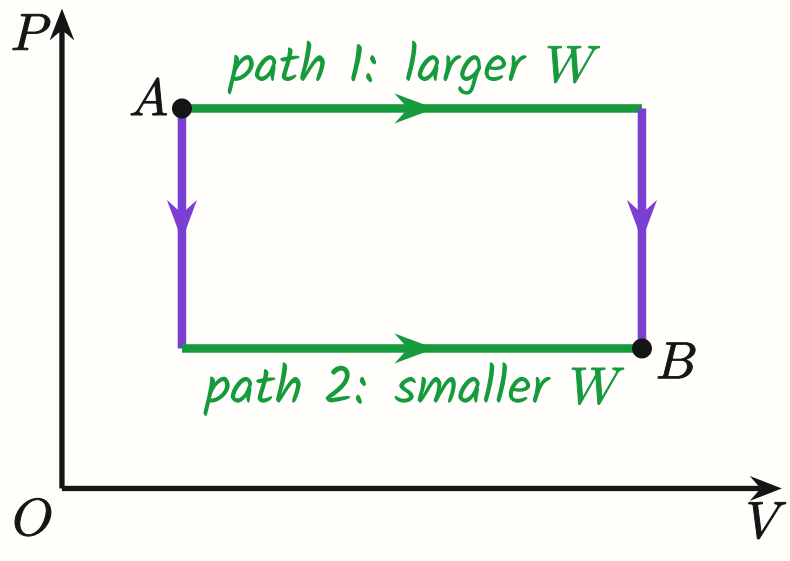
Work = Area Under P–V

1 Quasi-static process

- Infinitely SLOW: the gas is in equilibrium at every instant, so P, V, T are defined all the way and the path can be DRAWN on a P–V graph.
- Fast processes (free expansion, a burst) pass through non-equilibrium states: NO path on the graph.

2 Area and path dependence





W = area between the path and the V-axis

P on the vertical axis, V on the horizontal · V increasing: $W > 0$; V decreasing: $W < 0$; V constant: $W = 0$

- Same end states A, B: same ΔU, but different W, so different $Q = \Delta U + W$.

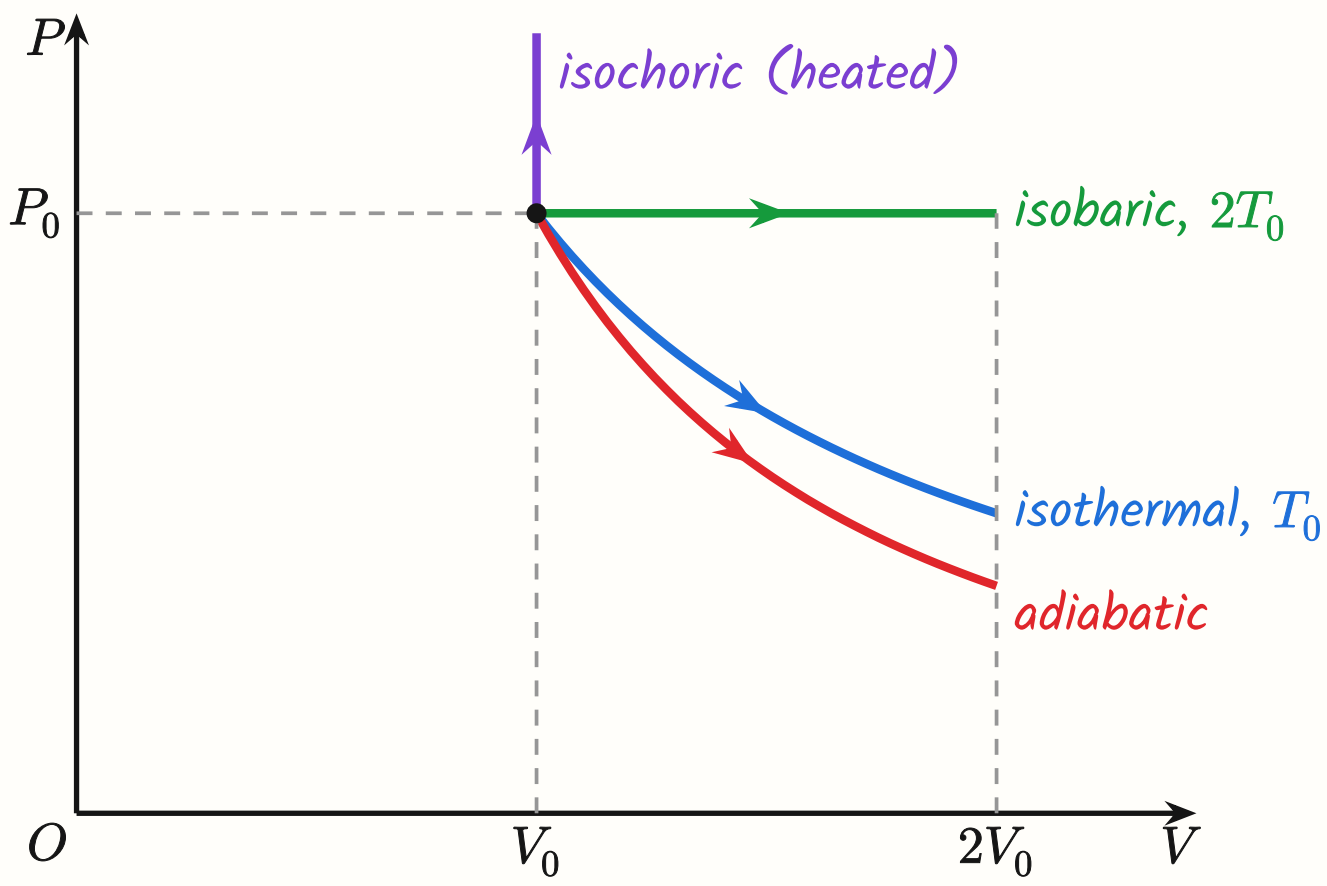
Example

Straight line from $(100\,\mathrm{kPa},\ 2\,\mathrm{L})$ to $(300\,\mathrm{kPa},\ 6\,\mathrm{L})$: trapezium, $W = \frac{1}{2}(100 + 300) \times (6 - 2)\,\mathrm{kPa\,L} = 800\,\mathrm{J}$ (since $1\,\mathrm{kPa\,L} = 1\,\mathrm{J}$).

Processes on One P–V Graph

same start (V_0, P_0), expansion to $2V_0$, quasi-static, ideal gas, $\gamma = 1.4$

1 Four processes from one state



$$W_{isobaric} > W_{isothermal} > W_{adiabatic}$$

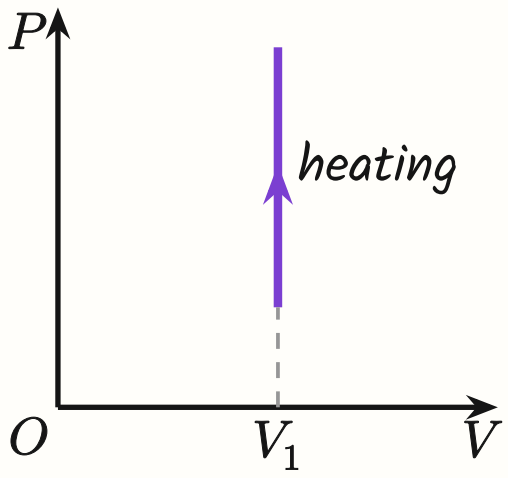
SAME initial state, SAME final volume $2V_0$ (expansion) · order of the areas under the curves

2 The numbers (from the curves)

- *Isobaric:* $W = P_0V_0$, $T \to 2T_0$.
- *Isothermal:* $W = P_0V_0 \ln 2 = 0.693\, P_0V_0$, $T = T_0$.
- *Adiabatic:* $W = \dfrac{P_0V_0\,(1 - 2^{-0.4})}{0.4} = 0.605\, P_0V_0$,
  $T \to 2^{-0.4}\, T_0 = 0.758\, T_0$.
- *Isochoric:* $W = 0$ (vertical line, no area).

Isochoric and Isobaric

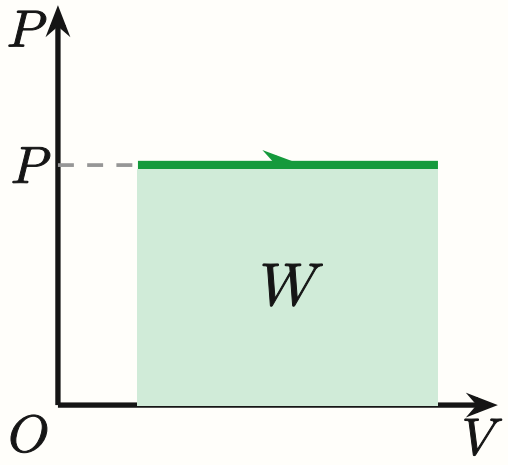
1 Isochoric (V constant)



$$W = 0, \qquad Q = \Delta U = nC_V \Delta T$$

RIGID container · P/T = constant (ideal gas) · all heat goes into U

2 Isobaric (P constant)



$$W = P\Delta V = nR\,\Delta T$$

$$Q = nC_P \Delta T, \quad \Delta U = nC_V \Delta T$$

ideal gas, constant pressure (free piston) · V/T = constant

$$\frac{\Delta U}{Q} = \frac{1}{\gamma} \qquad \frac{W}{Q} = 1 - \frac{1}{\gamma}$$

isobaric, ideal gas · monatomic: $\frac{3}{5}$ and $\frac{2}{5}$ · diatomic (rigid): $\frac{5}{7}$ and $\frac{2}{7}$

Example

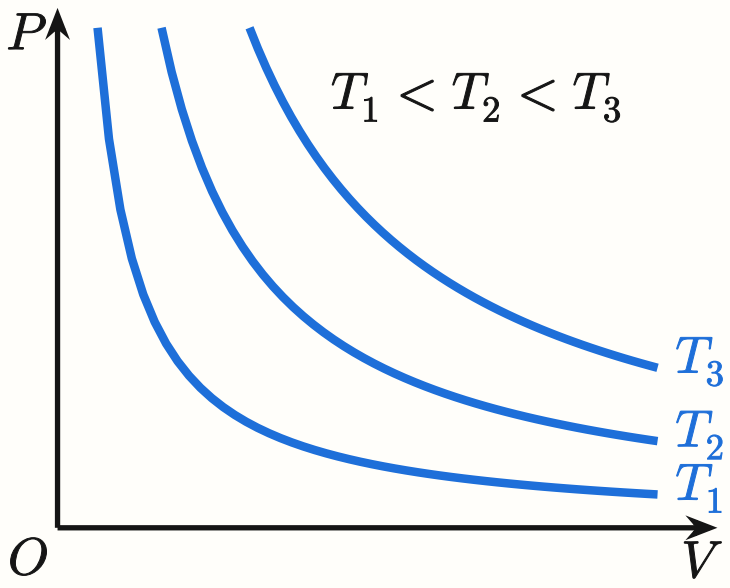
$2\,\text{mol}$ *monatomic gas heated by* $50\,\text{K}$ *at constant* P*:*

$Q = 2 \cdot \frac{5}{2}R \cdot 50 = 250R = 2078.5\,\text{J};$

$W = 2R \cdot 50 = 100R = 831.4\,\text{J};\ \Delta U = 150R = 1247.1\,\text{J}.$

Isothermal Process

1 T constant



$$PV = \text{constant}$$

ideal gas, n fixed, SLOW, in contact with a reservoir at T · hotter isotherm: farther from the origin

2 Work, heat, energy

$$W = nRT\ln\frac{V_2}{V_1} = nRT\ln\frac{P_1}{P_2}$$

$$\Delta U = 0 \qquad Q = W$$

ideal gas, reversible (quasi-static) isothermal · ln = natural log · expansion: heat IN = work OUT

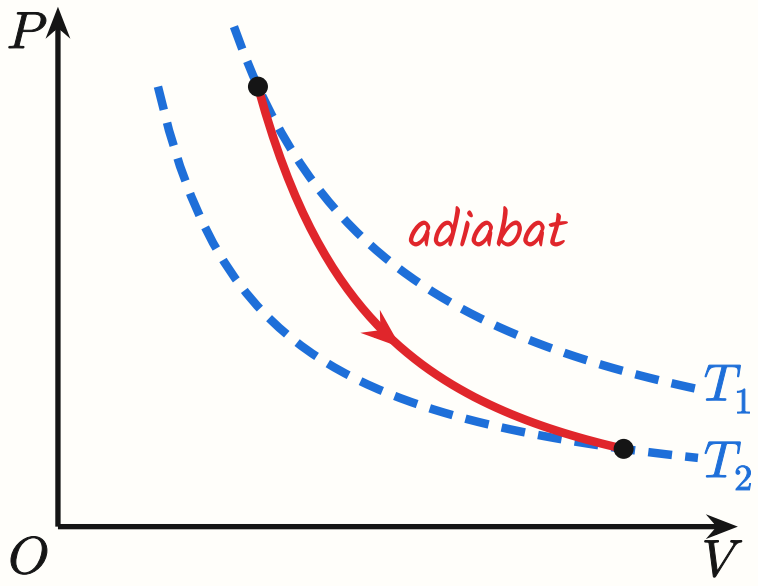
- *Molar heat capacity $C = \infty$: heat flows with NO change of T.*
- *Isothermal bulk modulus $B_T = -V\dfrac{dP}{dV} = P$.*

Example

$1\,\text{mol}$ *at* $300\,\text{K}$ *doubles its volume isothermally:*
$W = 8.314 \times 300 \times \ln 2 = 1729\,\text{J} = Q$ $(\ln 2 = 0.693)$.

Adiabatic Process

1 $Q = 0$



$$PV^{\gamma} = \text{const}$$

$$TV^{\gamma-1} = \text{const}$$

$$P^{1-\gamma}T^{\gamma} = \text{const}$$

ideal gas, REVERSIBLE (quasi-static) adiabatic, γ constant

- Adiabatic = NO heat exchange: perfectly insulated walls, or so fast that heat has no time to flow.
- From $dQ = 0$: $nC_V dT = -P dV$, with $PV = nRT$ gives $\dfrac{dT}{T} = -(\gamma - 1)\dfrac{dV}{V}$.

2 Work

$$W = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} = -\Delta U$$

reversible adiabatic, ideal gas · expansion: gas COOLS; compression: gas HEATS · $C = 0$

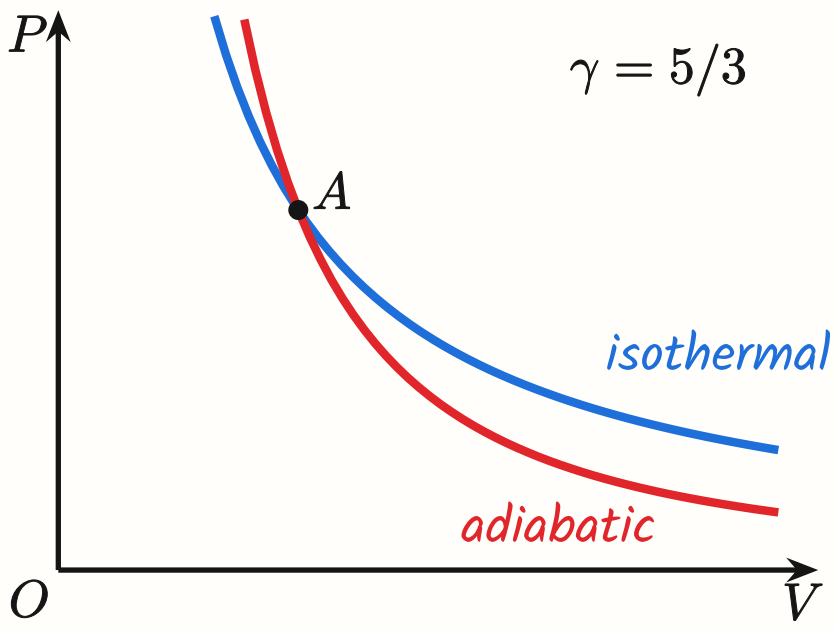
Example

Monatomic ($\gamma = 5/3$) gas at $250\,\mathrm{K}$ compressed reversibly and adiabatically to $1/8$ of its volume: $T_2 = 250 \times 8^{2/3} = 1000\,\mathrm{K}$, $P_2 = 8^{5/3}P_1 = 32P_1$.

Isothermal vs Adiabatic Slopes

ideal gas, quasi-static, both curves through the same state

1 Which curve is steeper?



$$\left(\frac{dP}{dV}\right)_T = -\frac{P}{V}$$

$$\left(\frac{dP}{dV}\right)_{ad} = -\gamma\frac{P}{V}$$

at the SAME point: adiabatic slope $= \gamma\times$ isothermal slope (steeper, as $\gamma > 1$)

2 Bulk modulus of a gas

$$B_T = P \qquad B_{ad} = \gamma P$$

$B = -V\dfrac{dP}{dV}$ · *ideal gas, quasi-static* · *sound in a gas is adiabatic:* $v = \sqrt{\gamma P/\rho}$

Example

At $P = 1.0 \times 10^5\ \text{Pa}$, $\gamma = 1.4$: $B_T = 1.0 \times 10^5\ \text{Pa}$, $B_{ad} = 1.4 \times 10^5\ \text{Pa}$.

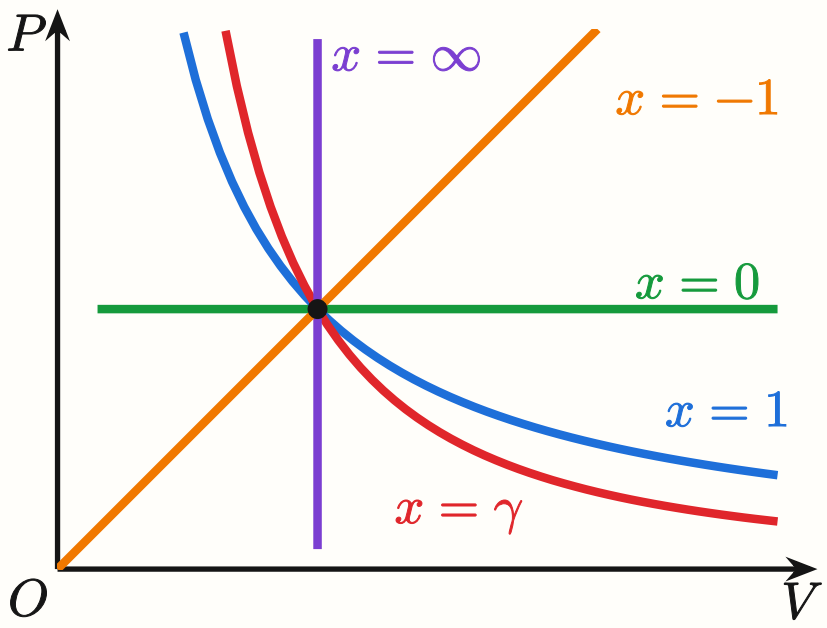
Golden points

- ★ *Expansion from one state to the same final V: adiabatic ends at LOWER P and does LESS work than isothermal.*
- ★ *Compression to the same final V: adiabatic ends at HIGHER P and needs MORE work.*

Polytropic Process

ideal gas, quasi-static, n fixed, $x \neq 1$ in the formulas

1 $PV^x = \text{constant}$



$$W = \frac{nR(T_1 - T_2)}{x - 1}$$

$$C = C_V + \frac{R}{1 - x}$$

C: molar heat capacity in the process.

$Q = nC\,\Delta T$

2 Special cases

- $x = 0$: isobaric, $C = C_P$. $x = 1$: isothermal, $C = \infty$.
- $x = \gamma$: adiabatic, $C = 0$. $x \to \infty$: isochoric, $C = C_V$.
- $1 < x < \gamma$: $C < 0$ (gas takes in heat yet COOLS on expansion).

Example

Monatomic gas, $P \propto V$ ($x = -1$): $C = \frac{3}{2}R + \frac{R}{2} = 2R$. Monatomic, $PV^2 = \text{const}$: $C = \frac{3}{2}R - R = \frac{R}{2}$.

Golden points

★ $P \propto V$: straight line through the origin, $T \propto V^2$, so the gas HEATS as it expands; W = trapezium area.

★ $x = 1$ is the isothermal case: use $W = nRT\ln\frac{V_2}{V_1}$ (the formula above needs $x \neq 1$).

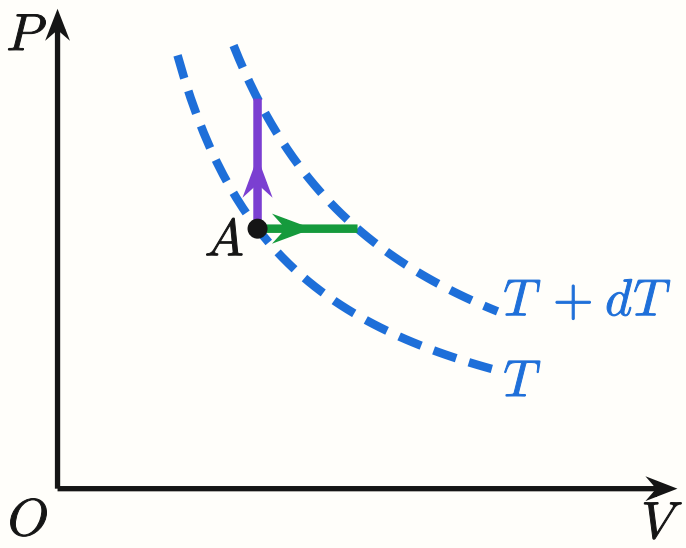
Specific Heats of a Gas

1 Definitions

$$C_V = \frac{1}{n}\left(\frac{dQ}{dT}\right)_V \qquad C_P = \frac{1}{n}\left(\frac{dQ}{dT}\right)_P$$

MOLAR heat capacities ($\mathrm{J\,mol^{-1}K^{-1}}$) · *per kg:* $c = Q/(m\,\Delta T)$ *in* $\mathrm{J\,kg^{-1}K^{-1}}$

2 Mayer's relation



$$C_P - C_V = R$$

IDEAL gas, molar values · constant P*:*
$nC_P dT = nC_V dT + P\,dV$ *and* $P\,dV = nR\,dT$

$$\gamma = \frac{C_P}{C_V} \qquad C_V = \frac{R}{\gamma - 1} \qquad C_P = \frac{\gamma R}{\gamma - 1}$$

ideal gas · $\gamma > 1$ *always*

- *Per kg:* $c_P - c_V = R/M$ (M *= molar mass in* $\mathrm{kg\,mol^{-1}}$). *A gas has many heat capacities:* C *depends on the PROCESS* ($-\infty$ *to* $+\infty$).

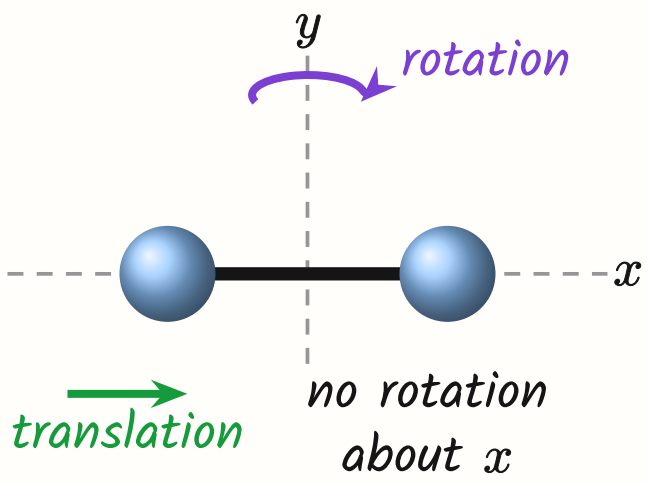
Example

Oxygen, $M = 0.032\,\mathrm{kg\,mol^{-1}}$:
$c_P - c_V = 8.314/0.032 = 259.8\,\mathrm{J\,kg^{-1}K^{-1}}$.

Degrees of Freedom and γ

equipartition: classical, thermal equilibrium; ideal gas

1 Equipartition of energy



$$U = \frac{f}{2} nRT \qquad C_V = \frac{f}{2} R$$

$$\gamma = 1 + \frac{2}{f}$$

$\frac{1}{2}k_B T$ per quadratic degree of freedom per molecule · ideal gas

2 Values (ordinary temperature)

Gas	f	C_V	C_P	γ
monatomic (He, Ar)	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} \approx 1.67$
diatomic, RIGID (O_2, N_2)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
diatomic + vibration (high T)	7	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{9}{7} \approx 1.29$
linear triatomic, rigid (CO_2)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
non-linear, rigid (H_2O, CH_4)	6	$3R$	$4R$	$\frac{4}{3} \approx 1.33$

- RIGID = no vibration: true at ordinary temperatures. One vibrational mode adds 2 (KE + PE) to f.

Golden points

★ More degrees of freedom ⇒ smaller γ: $1 < \gamma \leq \frac{5}{3}$ for an ideal gas.

Mixture of Gases

NON-REACTING ideal gases, rigid molecules

1 Heat capacities of a mixture

$$C_{V,mix} = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2} \qquad C_{P,mix} = C_{V,mix} + R$$

U adds up: $U = (n_1 C_{V1} + n_2 C_{V2})T$ · common temperature T

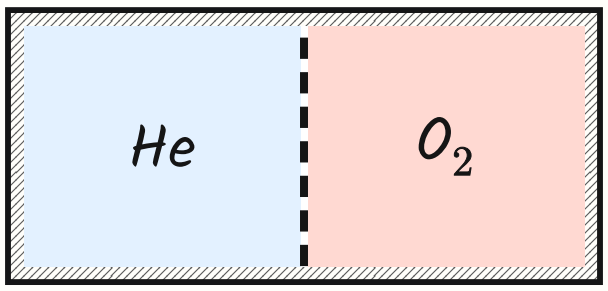
$$\frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

same condition · molar mass $M_{mix} = (n_1 M_1 + n_2 M_2)/(n_1 + n_2)$

Example

$1\,\text{mol}$ *He* + $1\,\text{mol}$ O_2: $C_V = (1.5R + 2.5R)/2 = 2R$, $C_P = 3R$, $\gamma_{mix} = 1.5$. *Check:* $2/(\gamma_{mix} - 1) = 1.5 + 2.5 = 4$.

2 Mixing in an insulated rigid box



partition removed

$$T_f = \frac{n_1 C_{V1} T_1 + n_2 C_{V2} T_2}{n_1 C_{V1} + n_2 C_{V2}}$$

$Q = 0$, $W = 0$ (rigid, insulated) $\Rightarrow$ total U conserved

Example

$1\,\text{mol}$ *He at* $300\,\text{K}$ + $1\,\text{mol}$ O_2 *at* $400\,\text{K}$:
$T_f = (1.5 \times 300 + 2.5 \times 400)/4 = 362.5\,\text{K}$.

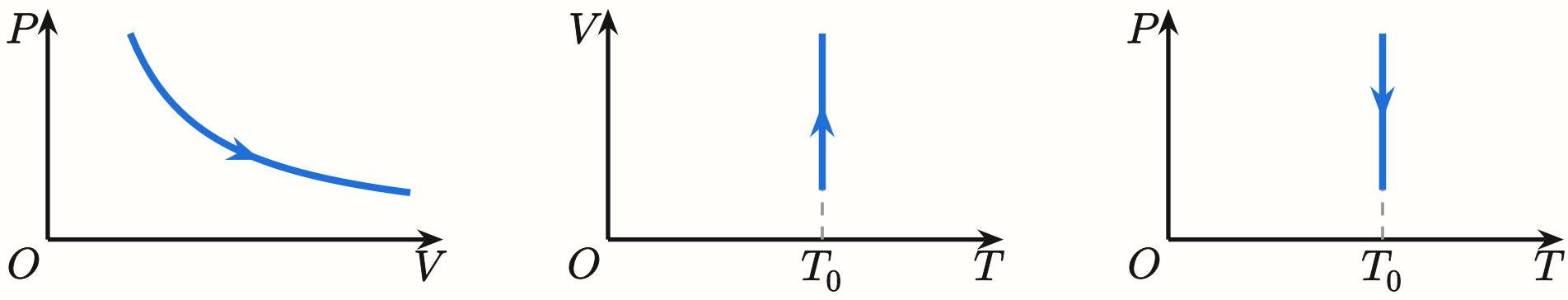
Golden points

★ γ_{mix} *is NOT the average of the γ's: here* $(\frac{5}{3} + \frac{7}{5})/2 = 1.533 \neq 1.5$.

Graphs: P–V, V–T, P–T (I)

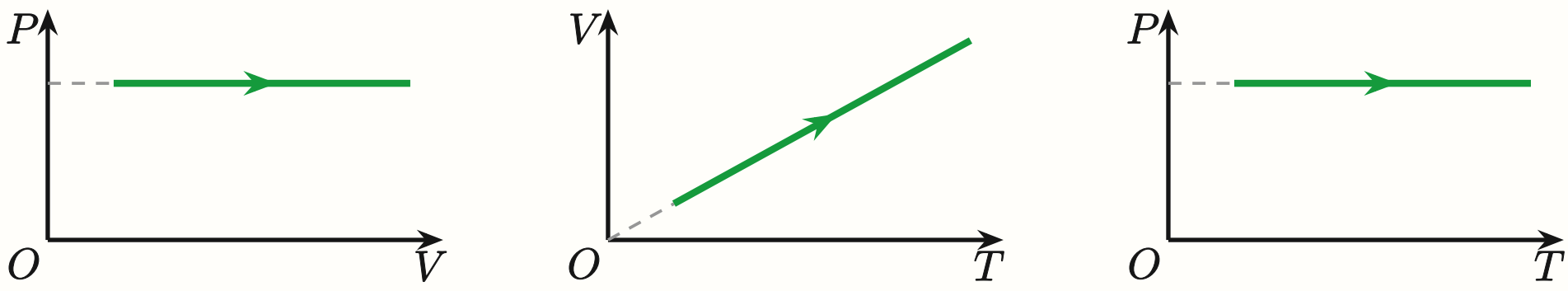
ideal gas, n fixed, quasi-static; T in kelvin

1 Isothermal (T constant)



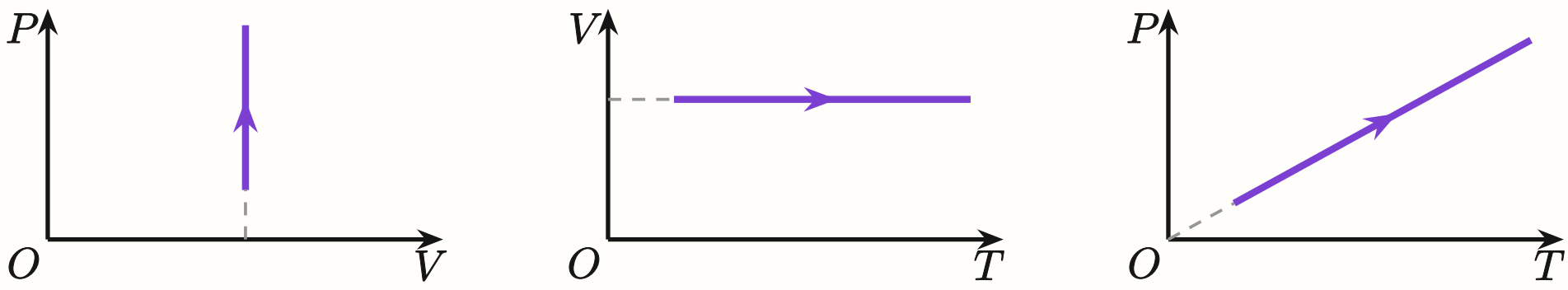
PV = const: hyperbola; vertical lines on V–T, P–T (expansion shown).

2 Isobaric (P constant)



$V \propto T$: line THROUGH THE ORIGIN on V–T; horizontal on the other two.

3 Isochoric (V constant)

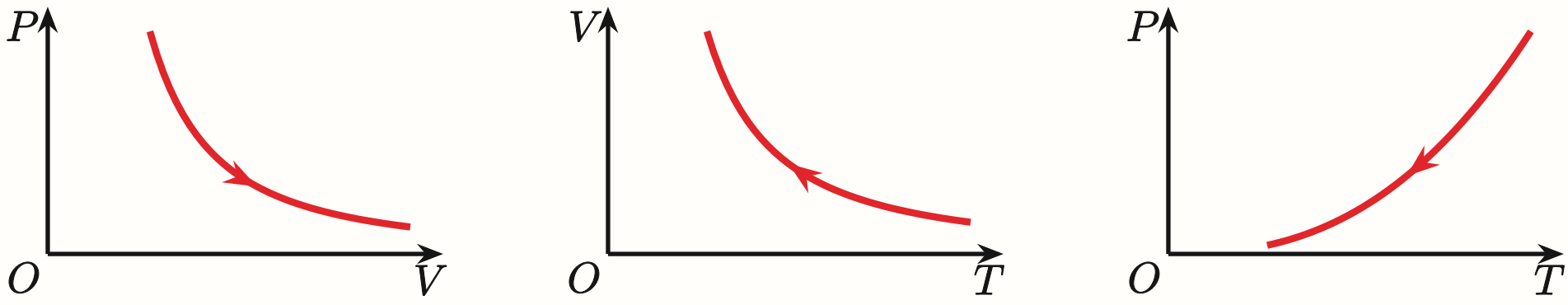


$P \propto T$: line THROUGH THE ORIGIN on P–T (rigid container, heating shown).

Graphs: P–V, V–T, P–T (2)

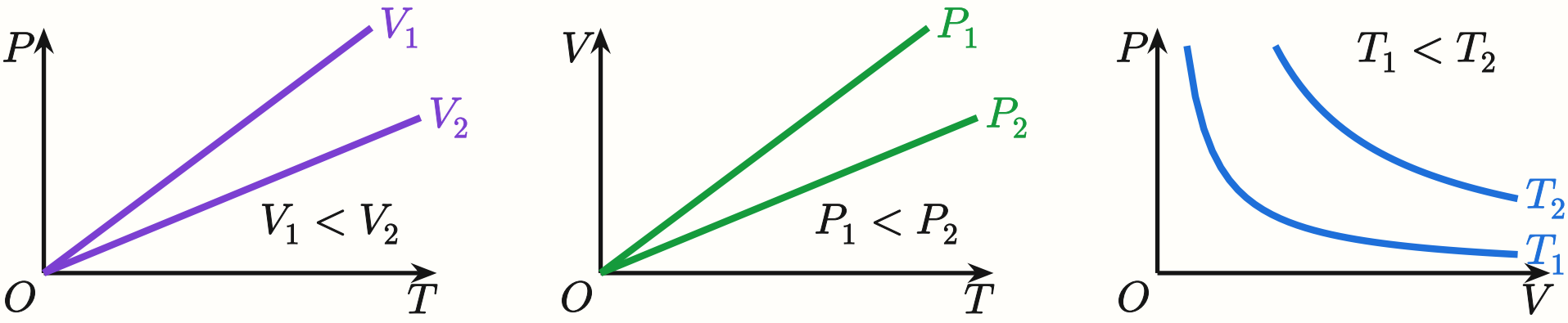
ideal gas, n fixed, quasi-static; T in kelvin

1 Reversible adiabatic (drawn for $\gamma = 5/3$)



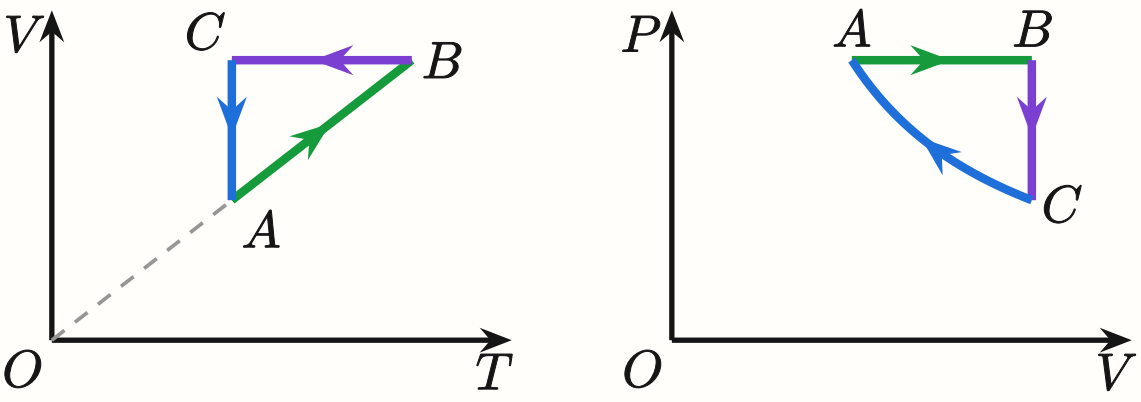
Expansion: $P \propto V^{-\gamma}$, $V \propto T^{-1/(\gamma-1)}$, $P \propto T^{\gamma/(\gamma-1)}$; T FALLS as V grows.

2 Reading the slopes



Steeper P–T line: SMALLER V. Steeper V–T line: SMALLER P. Farther isotherm: hotter.

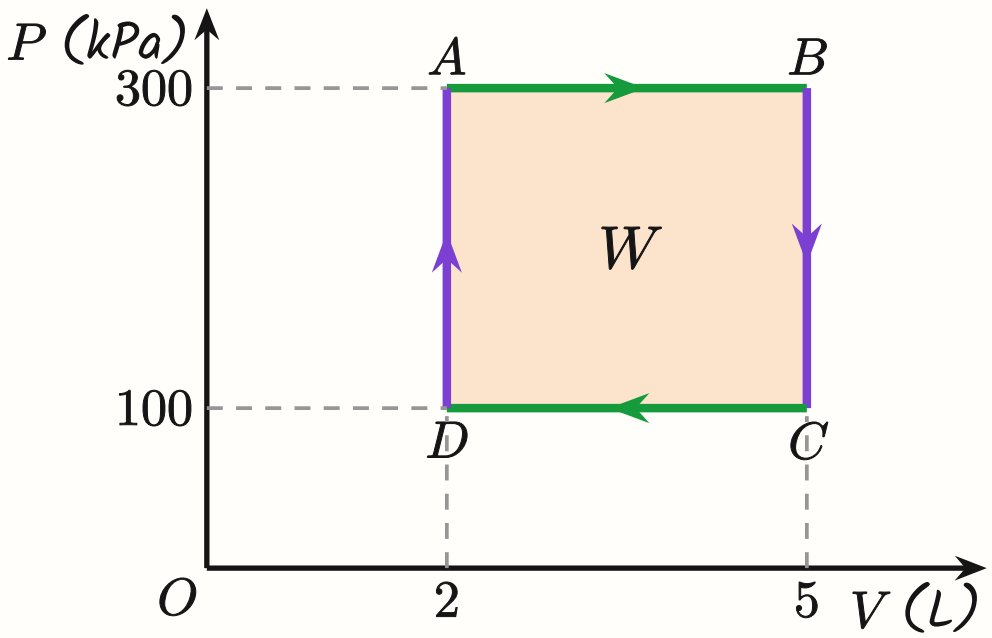
3 Converting a cycle: V–T to P–V



AB isobaric, BC isochoric, CA isothermal: CLOCKWISE on P–V, $W = P_0V_0(1 - \ln 2) = 0.307\,P_0V_0$.

Cyclic Process

1 Gas returns to its starting state



$$\Delta U_{cycle} = 0$$

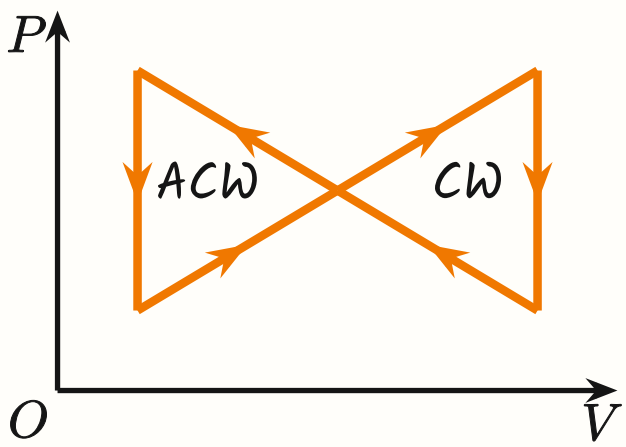
$$Q_{net} = W_{net}$$

W_{net} = ENCLOSED area · P up, V right: CLOCKWISE $W > 0$ (engine), ANTICLOCKWISE $W < 0$ (refrigerator)

Example

Cycle $ABCDA$ (clockwise):
$W = (300 - 100) \times (5 - 2)\,\text{kPa L} = 600\,\text{J} = Q_{net}$ (gas takes in $600\,\text{J}$ more than it gives out).

2 Figure-of-eight cycle



- Net work = (clockwise loop area) − (anticlockwise loop area).
- On a V–P graph (V vertical) the rule FLIPS; on P–T or V–T the area is NOT work: convert to P–V first.

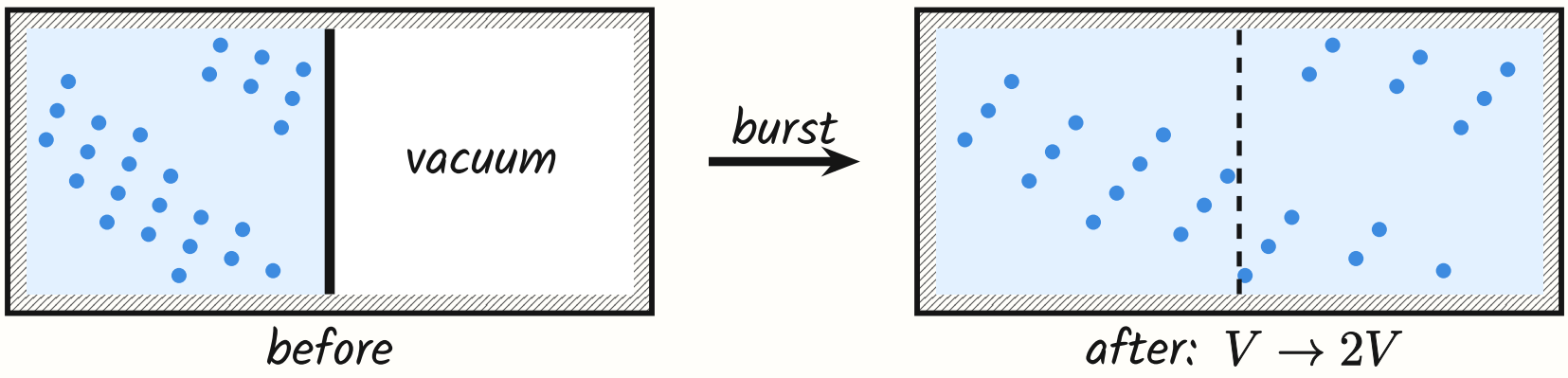
Golden points

★ $\Delta U = 0$ for the WHOLE cycle only; each step has its own $\Delta U = nC_V\Delta T$ (ideal gas).

★ $Q_{net} = W_{net}$, but heat ABSORBED (for η) exceeds Q_{net}.

Free Expansion

1 Gas rushes into vacuum



$$Q = 0, \quad W = 0 \;\Rightarrow\; \Delta U = 0$$

insulated, RIGID walls; expansion into VACUUM ($P_{ext} = 0$) · IDEAL gas: $T_f = T_i$, $P_f = P_i V_i / V_f$

- *IRREVERSIBLE and NOT quasi-static: no path on the P–V graph (only the two end points).*
- *$PV^{\gamma} =$ const does NOT apply although $Q = 0$ (it needs a reversible adiabatic).*
- *A real gas usually cools a little (work against intermolecular attraction).*

2 Against a constant external pressure

$$W = P_{ext}\,(V_2 - V_1)$$

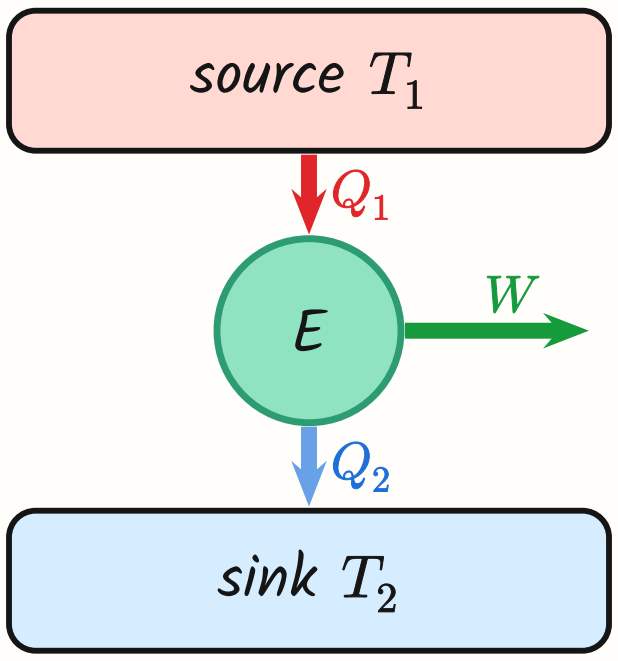
sudden expansion or compression against a FIXED P_{ext} (irreversible) · then $Q = \Delta U + W$ as usual

Example

Ideal gas, free expansion to twice the volume: T unchanged, P halves, $W = 0$, $Q = 0$.

Heat Engine

1 How it works



$$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

CYCLIC device ($\Delta U = 0$ per cycle) · Q_1: heat absorbed, Q_2: heat rejected (magnitudes) · $W = Q_1 - Q_2$

2 The cycle

- *Working substance (gas) absorbs Q_1 from the hot source, does work W, rejects Q_2 to the cold sink, returns to its initial state.*
- *$\eta < 1$ always: some heat MUST be rejected (Kelvin–Planck, p22).*

Example

Engine absorbs $1000\,\mathrm{J}$, *rejects* $600\,\mathrm{J}$ *per cycle:* $W = 400\,\mathrm{J}$, $\eta = 400/1000 = 40\%$.

Exam trap

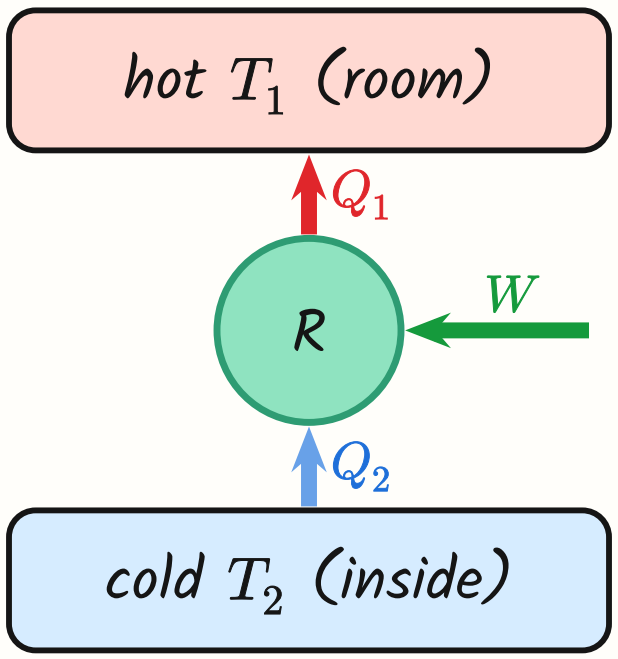
✗ *$\eta = W/Q_{net}$ (no: $\eta = W/Q_1$; Q_1 = sum of only the POSITIVE heats of the cycle).*

Golden points

★ *From a P–V cycle: W = enclosed area, Q_1 = add the heat of every step with $Q > 0$.*

Refrigerator and Heat Pump

1 Engine run in reverse



$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$$

coefficient of performance (COP) of a refrigerator · $Q_1 = Q_2 + W$ (cyclic, magnitudes)

2 Carnot (reversible) values

$$\beta_{Carnot} = \frac{T_2}{T_1 - T_2} \qquad COP_{pump} = \frac{Q_1}{W} = \beta + 1$$

reversible, between TWO reservoirs, T in kelvin · heat pump: the useful heat is Q_1

$$\beta = \frac{1 - \eta}{\eta}$$

the SAME reversible machine run as an engine (η) and as a refrigerator (β)

Example

Freezer at $250\,\mathrm{K}$, room at $300\,\mathrm{K}$, Carnot: $\beta = \frac{250}{50} = 5$. To remove $500\,\mathrm{J}$: $W = 100\,\mathrm{J}$, $Q_1 = 600\,\mathrm{J}$ to the room.

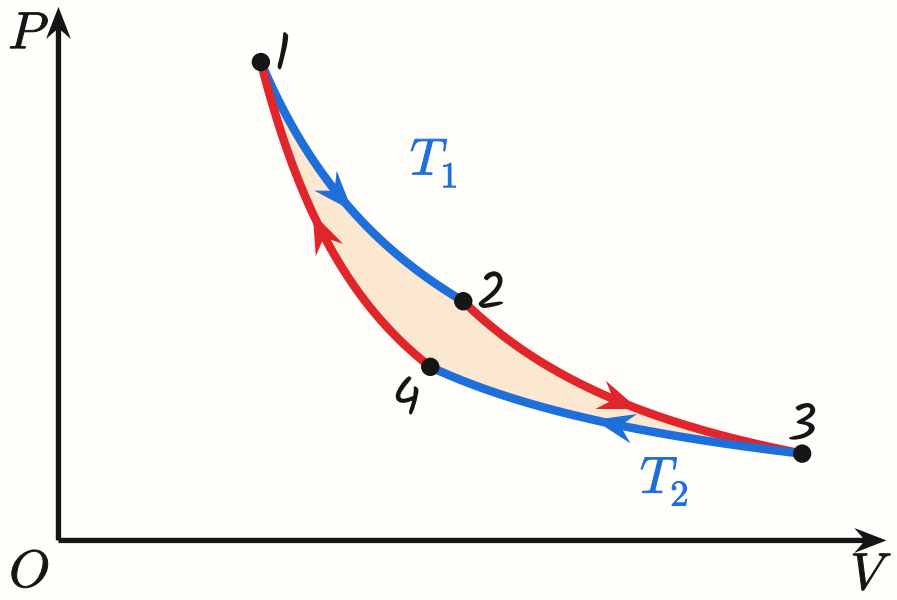
Golden points

★ β is NOT an efficiency: it can be (and usually is) more than 1.

Carnot Cycle

reversible, ideal gas, two reservoirs $T_1 > T_2$; drawn exactly for $\gamma = 5/3$, $T_1/T_2 = 1.5$

1 Two isotherms + two adiabats



- 1→2 isothermal at T_1: heat in $Q_1 = nRT_1 \ln(V_2/V_1)$
- 2→3 adiabatic: $T_1 \to T_2$
- 3→4 isothermal at T_2: heat out $Q_2 = nRT_2 \ln(V_3/V_4)$
- 4→1 adiabatic: $T_2 \to T_1$

2 Efficiency

$$\frac{V_2}{V_1} = \frac{V_3}{V_4} \Rightarrow \frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

$$\eta_{Carnot} = 1 - \frac{T_2}{T_1}$$

REVERSIBLE engine, TWO reservoirs, T in KELVIN · any working substance

Example

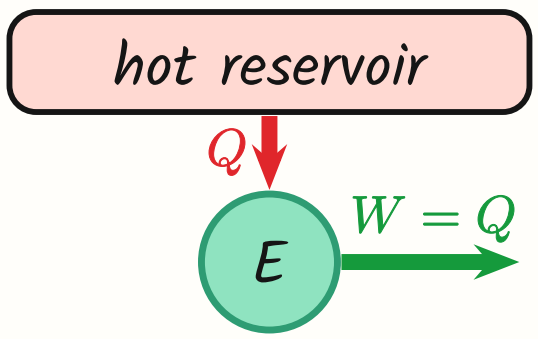
$T_1 = 500\,\mathrm{K}$, $T_2 = 300\,\mathrm{K}$: $\eta = 1 - \frac{300}{500} = 40\%$. *Drawn cycle* ($T_1/T_2 = 1.5$): $\eta = 1 - \frac{1}{1.5} = 33.3\%$.

Golden points

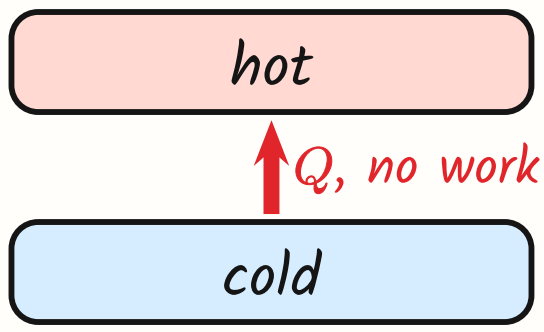
★ $\eta = 1$ *only if* $T_2 = 0\,\mathrm{K}$ *(unattainable): no engine is* 100% *efficient.*

Second Law of Thermodynamics

1 Two equivalent statements



IMPOSSIBLE: no sink



IMPOSSIBLE: no work input

- KELVIN–PLANCK (left): no CYCLIC process can have as its ONLY result heat taken from ONE reservoir and turned COMPLETELY into work ($\eta = 1$ impossible).
- CLAUSIUS (right): no process can have as its ONLY result heat flowing from a COLDER to a HOTTER body (a refrigerator needs work).

2 Carnot's theorem

$$\eta_{any} \leq \eta_{rev} = 1 - \frac{T_2}{T_1}$$

engines between the SAME two reservoirs T_1, T_2 · equality only for a REVERSIBLE engine

- (a) Nothing beats a reversible engine between two reservoirs.
  (b) ALL reversible ones have the SAME η.

Example

Claim: 30% between $400\,\mathrm{K}$ and $300\,\mathrm{K}$. Maximum $1 - \frac{300}{400} = 25\%$: IMPOSSIBLE.

Reversible and Irreversible

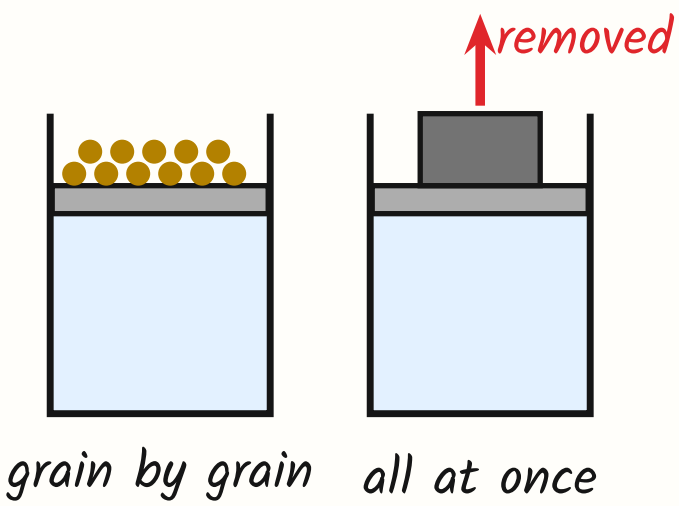
1 Reversible process

- Can be run BACKWARDS so that the system AND the surroundings both return to their initial states, with no change anywhere else.

$$\text{reversible} = \text{quasi-static} + \text{no dissipation}$$

no friction, viscosity or resistance heating · heat flows only across an INFINITESIMAL temperature difference

2 Irreversible process



- Left: nearly quasi-static (close to reversible). Right: sudden, IRREVERSIBLE.
- ALL natural processes are irreversible: free expansion, heat across a FINITE ΔT, friction, mixing, diffusion, sudden compression.

Golden points

★ Every reversible process is quasi-static; a quasi-static process WITH friction is still irreversible.

★ Reversible processes are ideal limits: between two reservoirs only a reversible engine reaches the Carnot efficiency.

Entropy

entropy basics (JEE Advanced); ideal gas, C_V constant

1 Definition

$$dS = \frac{dQ_{rev}}{T}$$

heat along a REVERSIBLE path, T in K · S is a STATE function, unit $\mathrm{J\,K^{-1}}$ · reservoir at T taking Q: $\Delta S = Q/T$

2 Ideal gas between two states

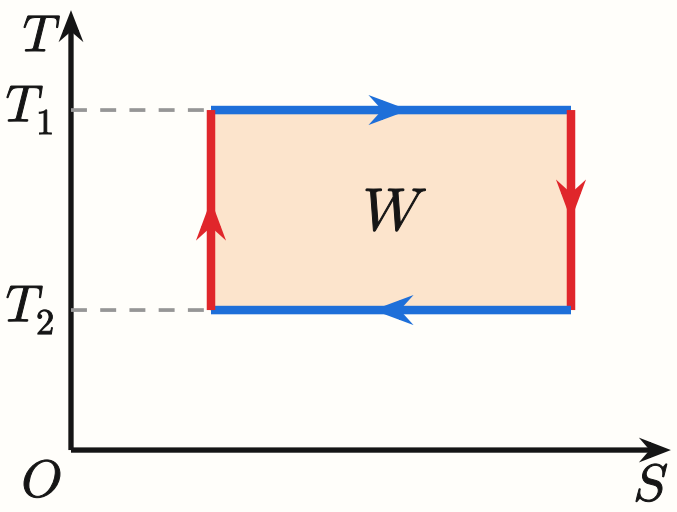
$$\Delta S = nC_V \ln\frac{T_2}{T_1} + nR\ln\frac{V_2}{V_1}$$

ANY process (even irreversible) between the two equilibrium states · isothermal: $nR\ln\frac{V_2}{V_1}$ · reversible adiabatic: $\Delta S = 0$

Example

$1\,\mathrm{mol}$ *free expansion to* $2V$*:*
$\Delta S = R\ln 2 = 8.314 \times 0.693 = 5.76\,\mathrm{J\,K^{-1}} > 0$ *although* $Q = 0$ *(an irreversible process).*

3 Second law and the T–S diagram



$$\Delta S_{universe} \geq 0$$

$= 0$ for a reversible process, > 0 for an irreversible one · Carnot cycle = RECTANGLE on T–S; its area = W (reversible cycle)

Process Summary Table

ideal gas, n moles, quasi-static unless stated; W = work BY the gas

1 One table, every process

Process	Law	W	Q	ΔU	C
isochoric	$\frac{P}{T} = \text{const}$	0	$nC_V\Delta T$	$nC_V\Delta T$	C_V
isobaric	$\frac{V}{T} = \text{const}$	$nR\Delta T$	$nC_P\Delta T$	$nC_V\Delta T$	C_P
isothermal	$PV = \text{const}$	$nRT\ln\frac{V_2}{V_1}$	$= W$	0	∞
adiabatic (reversible)	$PV^\gamma = \text{const}$	$nR(T_1-T_2)/(\gamma-1)$	0	$-W$	0
polytropic	$PV^x = \text{const}$	$nR(T_1-T_2)/(x-1)$	$nC\Delta T$	$nC_V\Delta T$	$C_V+\frac{R}{1-x}$
free expansion	no path	0	0	0	–
cyclic	closed loop	area	$= W$	0	–

2 How to use it

- Step 1: name the process (what is constant?). Step 2: $\Delta U = nC_V\Delta T$ ALWAYS (ideal gas). Step 3: W from the table or the area. Step 4: $Q = \Delta U + W$.
- Isothermal and polytropic work formulas need a QUASI-STATIC path; free expansion has no path, $W = 0$.

Golden points

★ $\Delta U = nC_V\Delta T$ even when the volume CHANGES (ideal gas, any process); $Q = nC_V\Delta T$ only at constant volume.

Standard Results

1 One-line results

Quantity	Result	Condition
First law	$Q = \Delta U + W$	W by the gas
Mayer	$C_P - C_V = R$	ideal gas, molar
γ	$1 + \frac{2}{f}$	ideal gas, equipartition
Isothermal W	$nRT \ln \frac{V_2}{V_1}$	ideal gas, reversible
Adiabatic	$PV^\gamma = \text{const}$	ideal gas, reversible
Slopes	adiabatic $= \gamma \times$ isothermal	same point on P–V
Bulk modulus	$B_T = P,\ B_{ad} = \gamma P$	ideal gas, quasi-static
Engine	$\eta = 1 - \frac{Q_2}{Q_1}$	any cyclic engine
Carnot	$\eta = 1 - \frac{T_2}{T_1}$	reversible, two reservoirs
Refrigerator	$\beta = \frac{T_2}{T_1 - T_2}$	Carnot (reversible)

2 Numbers to remember

- $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$, $\ln 2 = 0.693$, $1\,\mathrm{L\,atm} = 101.3\,\mathrm{J}$, $1\,\mathrm{kPa\,L} = 1\,\mathrm{J}$.
- γ (rigid): monatomic 1.67, diatomic 1.4, non-linear polyatomic 1.33.

Summary

$$Q = \Delta U + W \quad \bullet \quad PV^\gamma = \text{const} \quad \bullet \quad C_P - C_V = R \quad \bullet \quad \eta = 1 - \frac{T_2}{T_1}$$

ideal gas, W by the gas · PV^γ: reversible adiabatic · Carnot: reversible, two reservoirs

Common Traps

1 Exam traps

Avoid these

- ✗ Isothermal = no heat exchange (no: $\Delta T = 0$ but $Q = W \neq 0$; NO HEAT is ADIABATIC).
- ✗ $PV^{\gamma} =$ const in free expansion (only REVERSIBLE adiabatic; free expansion: T constant for an ideal gas).
- ✗ $\Delta U = nC_V\Delta T$ only at constant volume (no: ANY process of an ideal gas).
- ✗ Using T in °C in $\eta = 1 - T_2/T_1$ (always KELVIN).
- ✗ Taking W as work ON the gas with $Q = \Delta U + W$ (here W is work BY the gas).
- ✗ γ_{mix} = mean of the γ's (no: add $n/(\gamma - 1)$, that is, add nC_V).
- ✗ $\gamma = 1.4$ for every diatomic gas (only RIGID, no vibration: ordinary T).
- ✗ Clockwise = positive work on ANY graph (only on P–V with P vertical).
- ✗ $\eta = W/Q_{net}$ for a cycle (use Q_1 = heat ABSORBED only).
- ✗ COP cannot exceed 1 (it usually does: $\beta = T_2/(T_1 - T_2)$ for Carnot).
- ✗ Adiabatic slope $= \gamma$ (no: $\gamma\times$ isothermal slope at the SAME point).

Golden points

- ★ First ask: ideal gas? quasi-static/reversible? what is constant? which sign convention for W?

Questions: JEE Main Level (I)

ideal gas, $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$, W = work done BY the gas

Q1 · JEE Main level · MCQ

$2\,\mathrm{mol}$ of an ideal gas expand reversibly and isothermally at $300\,\mathrm{K}$ from $10\,\mathrm{L}$ to $20\,\mathrm{L}$. Work done by the gas: (A) $3.46\,\mathrm{kJ}$ (B) $1.73\,\mathrm{kJ}$ (C) $6.92\,\mathrm{kJ}$ (D) zero

Solution: $W = nRT\ln\frac{V_2}{V_1} = 2 \times 8.314 \times 300 \times \ln 2 = 3458\,\mathrm{J}$.

Answer: (A)

Q2 · JEE Main level · Numerical

A monatomic ideal gas absorbs $100\,\mathrm{J}$ of heat at constant pressure. Work done by the gas (in J)?

Solution: $W = Q\left(1 - \frac{1}{\gamma}\right) = 100 \times \left(1 - \frac{3}{5}\right) = 40\,\mathrm{J}$.

Answer: 40

Q3 · JEE Main level · MCQ

A Carnot engine has $\eta = 40\%$ with its sink at $300\,\mathrm{K}$. To make $\eta = 50\%$ with the same sink, the source temperature must rise by: (A) $50\,\mathrm{K}$ (B) $100\,\mathrm{K}$ (C) $150\,\mathrm{K}$ (D) $250\,\mathrm{K}$

Solution: $T_1 = \frac{300}{1-0.4} = 500\,\mathrm{K}$; new $T_1 = \frac{300}{1-0.5} = 600\,\mathrm{K}$; rise $100\,\mathrm{K}$.

Answer: (B)

Q4 · JEE Main level · Numerical

A diatomic ideal gas (rigid, $\gamma = 1.4$) at $27^\circ\mathrm{C}$ is compressed reversibly and adiabatically to $\frac{1}{32}$ of its volume. Final temperature in K?

Solution: $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = 300 \times 32^{0.4} = 300 \times 4 = 1200\,\mathrm{K}$.

Answer: 1200

Questions: JEE Main Level (2)

ideal gases, rigid molecules, quasi-static paths

Q5 · JEE Main level · MCQ

$2\,\text{mol}$ *He* + $1\,\text{mol}$ N_2 *(rigid).* γ_{mix}*:* *(A)* $\frac{17}{11}$ *(B)* $\frac{3}{2}$ *(C)* $\frac{23}{15}$ *(D)* $\frac{35}{23}$

Solution: $\frac{3}{\gamma-1} = \frac{2}{2/3} + \frac{1}{2/5} = 3 + 2.5 = 5.5$, *so*

$\gamma = 1 + \frac{6}{11} = \frac{17}{11} \approx 1.545.$

Answer: (A)

Q6 · JEE Main level · MCQ

Cycle $A(1\,\text{L}, 100\,\text{kPa}) \to B(3\,\text{L}, 100\,\text{kPa}) \to C(3\,\text{L}, 300\,\text{kPa}) \to A$*, straight lines. Net work BY the gas:* *(A)* $+200\,\text{J}$ *(B)* $-200\,\text{J}$ *(C)* $+400\,\text{J}$ *(D) zero*

Solution: *Area* $= \frac{1}{2} \times 2\,\text{L} \times 200\,\text{kPa} = 200\,\text{J}$*; ANTICLOCKWISE on* P*–*V*, so* $W = -200\,\text{J}.$

Answer: (B)

Q7 · JEE Main level · Numerical

A Carnot refrigerator works between $270\,\text{K}$ *(inside) and* $300\,\text{K}$ *(room). Work (in J) to remove* $900\,\text{J}$ *from inside?*

Solution: $\beta = \frac{270}{300-270} = 9$*;* $W = \frac{900}{9} = 100\,\text{J}.$

Answer: 100

Q8 · JEE Main level · MCQ

Ideal gas, same state, $V \to 2V$*: isobaric* W_1*, isothermal* W_2*, reversible adiabatic* W_3*.* *(A)* $W_1 > W_2 > W_3$ *(B)* $W_3 > W_2 > W_1$ *(C)* $W_2 > W_1 > W_3$ *(D) all equal*

Solution: Areas (p6): isobaric stays highest, adiabatic falls fastest.

Answer: (A)

Questions: JEE Advanced Level (I)

ideal gas, $n = 1\,\text{mol}$, quasi-static paths

A1 · JEE Advanced level · Multi-correct

$1\,\text{mol}$ of an ideal gas goes along a STRAIGHT line on the P–V graph from $A(V_0, 2P_0)$ to $B(2V_0, P_0)$. Then: (A) $T_A = T_B$ (B) the highest temperature on the path is $\frac{9P_0V_0}{4R}$ (C) work done by the gas is $\frac{3}{2}P_0V_0$ (D) the gas absorbs a net heat of $3P_0V_0$

Solution: $T_A = T_B = 2P_0V_0/R$ so $\Delta U = 0$. Line: $P = 3P_0 - P_0V/V_0$ gives $RT = 3P_0V - P_0V^2/V_0$, maximum at $V = 1.5V_0$: $T_{max} = \frac{9P_0V_0}{4R}$. Trapezium: $W = \frac{1}{2}(2P_0 + P_0)V_0 = \frac{3}{2}P_0V_0 = Q$.

Answer: (A), (B), (C)

A2 · JEE Advanced level · Numerical

$2\,\text{mol}$ of a monatomic ideal gas follow $PV^2 = \text{const}$ while cooling from $400\,\text{K}$ to $300\,\text{K}$. (i) Molar heat capacity in units of R? (ii) Heat absorbed (in J)?

Solution: (i) $C = C_V + \frac{R}{1-x} = \frac{3}{2}R - R = 0.5R$. (ii) $Q = nC\Delta T = 2 \times 0.5R \times (-100) = -100R = -831.4\,\text{J}$: the gas LOSES $831.4\,\text{J}$ (it expands, as $TV = \text{const}$).

Answer: (i) 0.5 (ii) −831.4

Golden points

- ★ Straight line $P = a - bV$: $T \propto PV$ is MAXIMUM at $V = a/(2b)$ (half the V-intercept), if that point lies on the segment.
- ★ Polytropic heat: $Q = nC\Delta T$ with $C = C_V + \frac{R}{1-x}$; a negative C means heat out while T rises, or heat in while T falls.

Questions: JEE Advanced Level (2)

ideal gas, $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$

A3 · JEE Advanced level · Multi-correct

Engines work between reservoirs at $500\,\mathrm{K}$ and $300\,\mathrm{K}$, each taking $1000\,\mathrm{J}$ per cycle from the hot one. Which are POSSIBLE? (A) a reversible engine giving $W = 400\,\mathrm{J}$ (B) an engine giving $W = 450\,\mathrm{J}$ (C) an engine giving $W = 300\,\mathrm{J}$ and rejecting $700\,\mathrm{J}$ (D) an engine for which $\Delta S_{universe} < 0$

Solution: $\eta_{max} = 1 - \frac{300}{500} = 0.4$: (A) is the Carnot limit, (B) needs $45\% > 40\%$: impossible. (C)

$\Delta S_{universe} = -\frac{1000}{500} + \frac{700}{300} = +0.333\,\mathrm{J\,K^{-1}} > 0$: possible. (D) violates the second law.

Answer: (A), (C)

A4 · JEE Advanced level · Numerical

$1\,\mathrm{mol}$ of a diatomic ideal gas (rigid, $\gamma = 1.4$) at $300\,\mathrm{K}$ in an insulated vessel expands FREELY into vacuum to twice its volume, then is compressed reversibly and adiabatically back to the original volume. Work done ON the gas (in J)?

Solution: Free expansion: $Q = 0$, $W = 0$, T stays $300\,\mathrm{K}$. Reversible adiabatic: $T_f = 300 \times 2^{0.4} = 395.85\,\mathrm{K}$.

$W_{on} = \Delta U = \frac{5}{2}R \times (395.85 - 300) = 1992\,\mathrm{J}$.

Answer: ≈ 1992

Golden points

- ★ Carnot test: $W \le Q_1(1 - T_2/T_1)$, that is, $\Delta S_{universe} \ge 0$.
- ★ Free expansion keeps T (ideal gas) but RAISES entropy; only the reversible step obeys $TV^{\gamma-1} = \text{const}$.

Questions: JEE Advanced Level (3)

ideal gas, quasi-static, W = work done BY the gas

A5 · JEE Advanced level · Matching

List I: (P) isobaric expansion (Q) isothermal expansion (R) reversible adiabatic expansion (S) isochoric cooling. List II: (1) $W = 0$, $Q = \Delta U < 0$ (2) $Q = 0$, $\Delta U < 0$ (3) $\Delta U = 0$, $Q = W > 0$ (4) Q, W, ΔU all > 0

Solution: (P) $W = nR\Delta T > 0$ and T rises; (Q) T constant; (R) $W = -\Delta U > 0$; (S) rigid, T falls.

Answer: P-4, Q-3, R-2, S-1

A6 · JEE Advanced level · Numerical

A monatomic ideal gas goes round the rectangle $(V_0, P_0) \to (V_0, 2P_0) \to (2V_0, 2P_0) \to (2V_0, P_0) \to (V_0, P_0)$. Efficiency of this cycle (in %)?

Solution: $W = P_0V_0$. Heat in: isochoric $\frac{3}{2}(2P_0V_0 - P_0V_0) = 1.5P_0V_0$, isobaric $\frac{5}{2}(4P_0V_0 - 2P_0V_0) = 5P_0V_0$; $Q_1 = 6.5P_0V_0$. $\eta = \frac{1}{6.5} = 15.4\%$ (a Carnot engine between the same extreme temperatures: $1 - \frac{1}{4} = 75\%$).

Answer: ≈ 15.4

Golden points

- ★ Cycle efficiency: Q_1 = heat of the steps where the gas takes heat in; W = enclosed area.
- ★ No cycle beats a Carnot engine working between its HIGHEST and LOWEST temperatures.